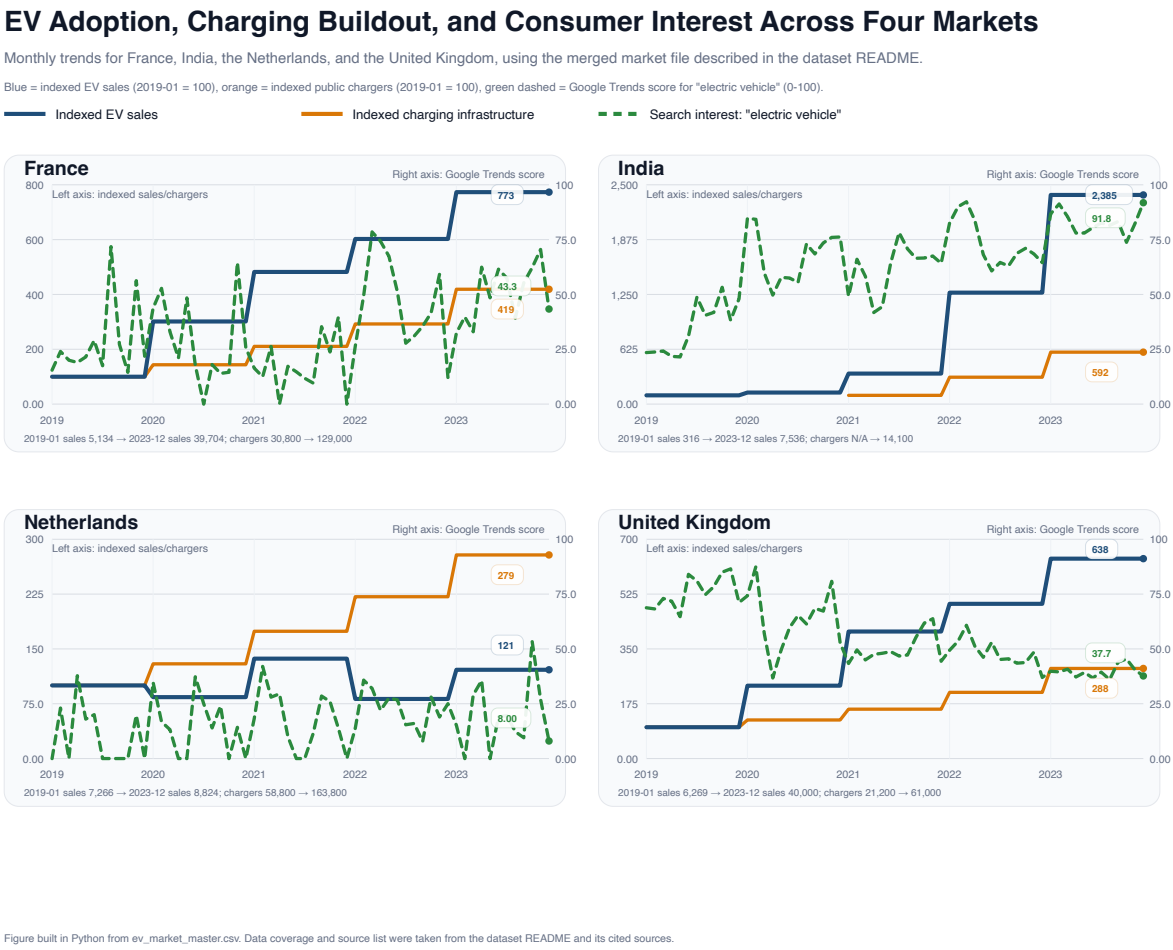
EV Adoption, Charging Buildout, and Consumer Interest Across Four Markets

Monthly trends for France, India, the Netherlands, and the United Kingdom, using the merged market file described in the dataset README.



Main figure produced in Python from `ev_market_master.csv`. The figure compares country-level EV adoption, public charging infrastructure, and Google Trends interest from 2019-01 to 2023-12.

Legend

> **Panels:** Each panel represents one country with a complete country-level monthly series in the merged market file.
>
> **Blue line:** Indexed monthly EV sales (`units_sold`), with 2019-01 set to 100 within each country.
>
> **Orange line:** Indexed cumulative public chargers (`total_chargers_cumulative`), also scaled to 2019-01 = 100.
>
> **Green dashed line:** Google Trends search interest for the keyword `electric vehicle`, shown on its original 0–100 scale.
>
> **Text labels:** Endpoint labels summarize the latest observed index or trend value in each panel.

Findings

- India shows the fastest relative EV expansion in the dataset: monthly sales rise from 316 in 2019-01 to 7,536 in 2023-12, a 23.85x increase.
- France and the United Kingdom finish 2023 at the largest monthly sales volumes (39,704 and 40,000), and both markets show very strong sales–charger co-movement (r = 0.968 and 0.962).
- The Netherlands has the largest charger base by 2023-12 (163,800), but its sales peak earlier (2021-01-01) and remain much flatter than France or the UK.
- Search interest is not a universal proxy for demand: the UK reaches 40,000 monthly sales in 2023-12 while "electric vehicle" search interest falls to 37.7.

Selected Computed Statistics

Country	Sales Growth (2019-01 to 2023-12)	End-2023 Sales	End-2023 Chargers	Sales-Chargers r	Sales-Search r	Sales-Gasoline r
France	7.73x	39,704	129,000	0.968	0.394	0.658
India	23.85x	7,536	14,100	0.999	0.634	0.662
Netherlands	1.21x	8,824	163,800	0.282	-0.004	-0.046
United Kingdom	6.38x	40,000	61,000	0.962	-0.84	0.739

These values were computed directly from the merged monthly market file and are included here so the report remains self-contained without needing a separate JSON appendix.

Data and Methods

- The analysis uses `ev_market_master.csv`, the merged dataset described in the local README.
- Only rows where `brand = All` were kept so the figure compares country-level EV demand rather than brand-specific sales.
- The main figure focuses on the four countries with complete `brand = All` monthly histories from 2019-01 to 2023-12: France, India, the Netherlands, and the United Kingdom.
- `units_sold` and `total_chargers_cumulative` were indexed to 2019-01 = 100 within each country so adoption and infrastructure growth can be compared on the same panel despite different raw scales.
- `trend_electric_vehicle` was overlaid as a dashed line on a separate 0–100 axis because Google Trends is already normalized.
- The script computes descriptive summaries and selected pairwise correlations only as supporting evidence for the visual findings; the main contribution of the project is the comparative visualization itself.

README-cited data sources:

- IEA Global EV Outlook 2024 (https://www.iea.org/reports/global-ev-outlook-2024)
- Google Trends via pytrends
- BYD Co. Ltd monthly sales press releases
- Tesla IR quarterly reports
- AFDC (US DOE Alternative Fuels Station Locator)
- IEA Charging Infrastructure data
- GlobalPetrolPrices.com / IEA Energy Prices

Significance Statement

Electric-vehicle adoption matters for climate policy, energy demand, and transportation planning. This figure is significant because it shows that EV demand is not driven by a single factor: charging buildout, consumer attention, and market context move together differently across countries. That comparison makes the visualization useful for discussing where infrastructure appears to support adoption and where demand may be shaped by broader policy or consumer conditions.

GitHub Archive Link

https://github.com/kevin0699/Data-mining-final-report

This repository archives the Python code, report assets, and dataset documentation for the final project.